

AI-Based Nutritional Assessment System for Deficiency Detection and Personalized Meal Planning

Technology Domain

Artificial Intelligence • Machine Learning • Full Stack Development • Healthcare Informatics

Team Size

6–7 Members

Development Methodology

Agile Scrum (6 Weeks)

1. Executive Summary

The AI-Based Nutritional Assessment System is an intelligent healthcare web application that predicts nutritional deficiencies using Machine Learning and generates personalized dietary recommendations based on user health information.

The application collects demographic information, dietary habits, symptoms, and optionally blood test values to estimate the likelihood of multiple nutritional deficiencies. Based on the prediction, the system creates a personalized meal plan, recommends nutrient-rich foods, tracks health progress, and provides interactive dashboards.

Unlike traditional calorie-tracking applications, this system focuses on **preventive healthcare**, helping users understand potential deficiencies before they become severe.

2. Project Objectives

- Detect nutritional deficiencies using Machine Learning.
- Build a comprehensive digital health profile for each user.
- Track daily food intake and nutrient consumption.
- Analyze symptoms and laboratory test values.
- Recommend personalized meal plans and recipes.
- Monitor health improvements through dashboards and progress tracking.
- Deliver a scalable, production-ready healthcare platform.

3. Project Scope

Included Features

- User Authentication
- Health Profile
- Food Diary
- Symptom Assessment
- Blood Report Upload
- AI-Based Deficiency Prediction
- Personalized Meal Planning
- Recipe Recommendation
- Nutrition Dashboard
- Progress Tracking
- Admin Dashboard

Future Scope

- OCR-based blood report extraction
- Food image recognition
- Voice-enabled nutrition assistant
- Wearable device integration
- Mobile application
- Doctor consultation portal

4. Functional Modules

Module 1 – Authentication & User Management

Features:

- User Registration

- Login (JWT Authentication)
 - Forgot Password
 - Role-Based Access (User/Admin)
 - User Profile
-

Module 2 – Health Assessment System

Features:

- Health Profile
 - BMI (Body Mass Index) Calculator
 - BMR (Basal Metabolic Rate) Calculator
 - Daily Calorie Requirement
 - Activity Level
 - Medical Conditions
 - Allergies
 - Dietary Preferences
 - Health Goals
-

Module 3 – Food Diary & Nutrition Tracking

Features:

- Daily Meal Logging
- Search Foods
- Barcode Support (OpenFoodFacts)
- Nutrient Calculator
- Water Intake Tracker
- Calorie Counter
- Macro & Micronutrient Tracking

Data Sources:

- USDA FoodData Central
 - IFCT (Indian Food Composition Tables)
 - OpenFoodFacts
-

Module 4 – Symptom Assessment

Questionnaire covering:

- Fatigue
 - Hair Loss
 - Muscle Weakness
 - Skin Conditions
 - Mood Changes
 - Bone Pain
 - Poor Vision
 - Slow Healing
 - Appetite Loss
 - Sleep Quality
-

Module 5 – Blood Report Module

Users can:

- Enter laboratory values manually
- Upload blood reports
- Extract values using OCR

Supported values:

- Hemoglobin
- Vitamin D

- Vitamin B12
- Iron
- Calcium
- Ferritin
- Blood Sugar

Module 6 – Machine Learning Engine

Dataset Sources

- | | | |
|-------------------------|---|-------------------------|
| • NHANES | - | For ML Training |
| • USDA FoodData Central | - | Food Nutrition Database |
| • IFCT | - | Indian Food Database |
| • Open Food Fact | - | World Food Search |

Data Processing

- Cleaning
- Normalization
- Feature Engineering
- Label Encoding
- Missing Value Handling

Models

- Random Forest

Prediction Targets:

- Iron Deficiency
- Vitamin D
- Vitamin B12
- Calcium
- Protein

- Folate
- Vitamin C
- Magnesium
- Zinc
- Potassium

Explainability

- SHAP Values
 - Feature Importance
-

Module 7 – Recommendation Engine

Based on predicted deficiencies, the engine generates:

Personalized Diet

- Breakfast
- Lunch
- Dinner
- Snacks

Nutritional Advice

- Foods to Eat
- Foods to Avoid
- Supplement Suggestions
- Hydration Goals

Recipe Sources:

- USDA FoodData Central API - (Provides - Calories, Vitamins, Minerals, Macronutrients)
- Spoonacular API - (Provides – Recipe, Meal planning, Nutrition, Ingredients, Food Images, Daily Meal Plan)
- Open Food Fact API - (Provides - food Lookup, Barcode Scanning)

AI-powered explanations generated using:

- OpenAI GPT-5.5 or Google Gemini
-

Module 8 – Health Dashboard

Dashboard includes:

- Nutrition Score
- Deficiency Risk Cards
- Daily Calories
- Water Intake
- Nutrient Trends
- Weekly Meal Plan
- Progress Graphs
- Recommendation History

Charts:

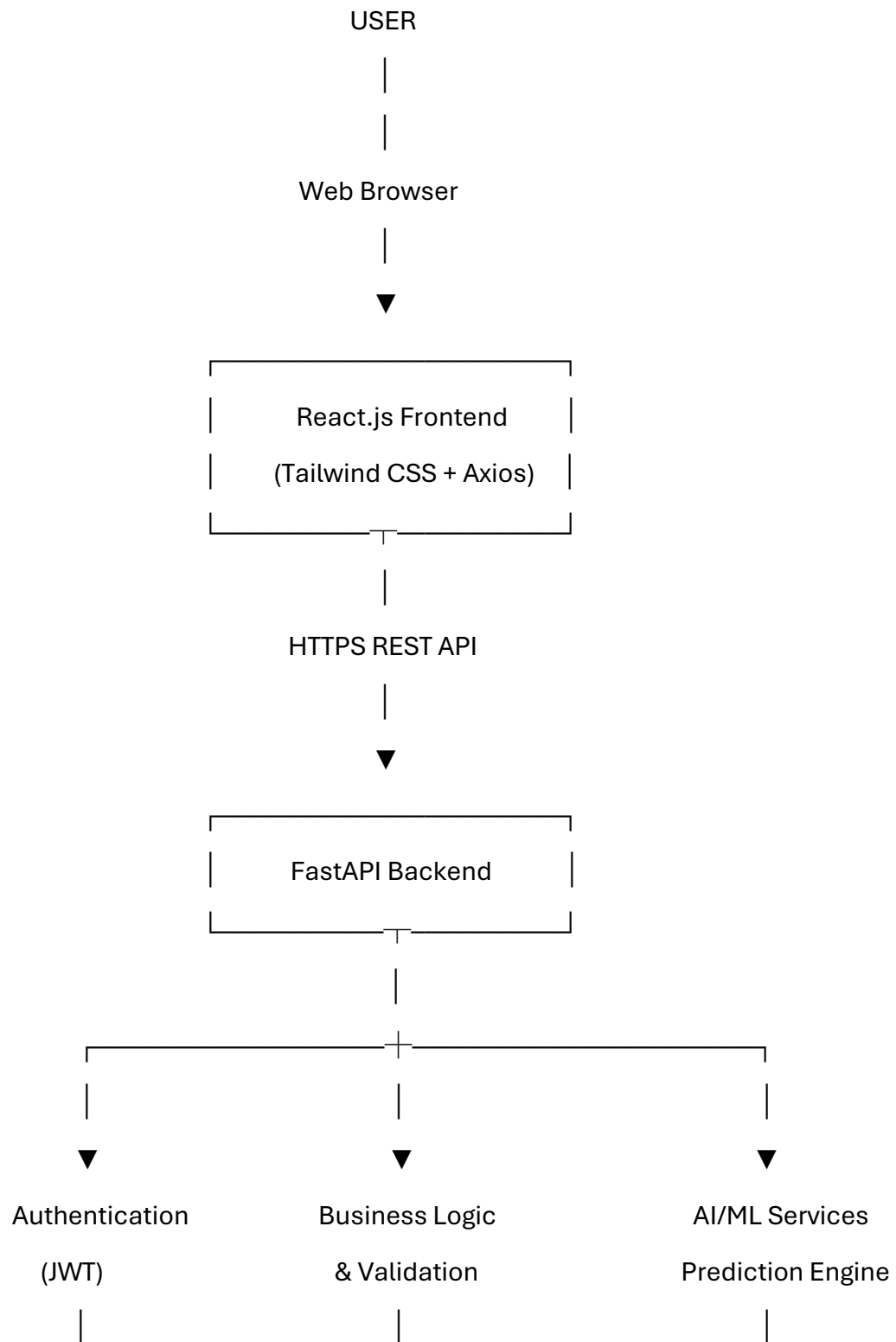
- Line Charts
 - Bar Charts
 - Pie Charts
-

Module 9 – Admin Panel

Admin Features:

- User Management
- Food Database Management
- Nutrition Dataset Management
- Analytics Dashboard
- Prediction Statistics
- Meal Database Management

5. System Architecture



- SHAP

APIs

- USDA FoodData Central
- OpenFoodFacts
- Spoonacular
- OpenAI GPT-5.5 / Google Gemini

Tools

- Git
- GitHub
- Docker
- VS Code
- Postman

7. GitHub Repository Structure

```
AI-Based-Nutritional-Assessment-System/  
|  
├── docs/  
├── frontend/  
├── backend/  
├── ml/  
├── database/  
├── tests/  
├── postman/  
├── deployment/  
├── scripts/  
├── .github/  
├── README.md  
└── docker-compose.yml
```

8. Database Tables

- Users

- Health Profiles
 - Food Diary
 - Symptoms
 - Blood Reports
 - Nutritional Intake
 - Predictions
 - Meal Plans
 - Recipes
 - Progress Logs
-

9. Frontend Pages

Public

- Landing Page
- About
- Login
- Register

User

- Dashboard
- Health Assessment
- Food Diary
- Blood Report
- Prediction Results
- Meal Planner
- Nutrition Tracker
- Reports
- Profile

- Settings

Admin

- Dashboard
 - Users
 - Food Database
 - Analytics
 - Prediction Reports
-

10. API Endpoints

Authentication

- POST /register
- POST /login

Profile

- GET /profile
- PUT /profile

Health Assessment

- POST /health-profile
- POST /food-diary
- POST /symptoms
- POST /blood-report

Prediction

- POST /predict

Meal Planner

- GET /meal-plan

Dashboard

- GET /dashboard

Admin

- CRUD operations for users, foods, and meal plans
-

11. Six-Week Development Plan

Week 1

Project setup, GitHub repository, authentication, database design, health profile.

Week 2

Food diary, symptom assessment, blood report module, nutrient calculation APIs.

Week 3

Dataset preparation, preprocessing, Random Forest training, prediction API.

Week 4

ML integration, SHAP explainability, recommendation engine, prediction history.

Week 5

Meal planner, recipe integration, nutrition dashboard, progress tracking, admin panel.

Week 6

Integration testing, deployment, optimization, documentation, presentation, and demo.

12. Testing Strategy

- Unit Testing
 - API Testing (Postman)
 - Integration Testing
 - Cross-Browser Testing
 - Performance Testing
 - User Acceptance Testing
-

13. Deployment

- **Frontend:** Vercel
 - **Backend:** Render
 - **Database:** Supabase PostgreSQL or Neon
 - **Version Control:** GitHub
 - **Containerization:** Docker
-

14. Final Deliverables

- React.js Web Application
- FastAPI REST Backend
- PostgreSQL Database
- Trained Machine Learning Models
- Personalized Meal Recommendation Engine
- Interactive Nutrition Dashboard
- Admin Panel
- API Documentation
- GitHub Repository
- Testing Report
- Deployment Guide
- Final Technical Report
- Project Presentation (PPT)
- Demo Video

15. User Account / User Profile Detailed Information

Purpose: Authentication & User Identification

Basic Details

- Full Name
 - Email Address
 - Password
 - Phone Number (Optional)
 - Profile Picture (Optional)
-

Personal Information

Purpose: Calculate nutritional requirements.

Field	Example
-------	---------

Age	22
-----	----

Gender	Male/Female/Other
--------	-------------------

Height	172 cm
--------	--------

Weight	68 kg
--------	-------

BMI	Auto Calculated
-----	-----------------

Occupation	Student, Office Worker, etc.
------------	------------------------------

Activity Level	Sedentary, Moderate, Active
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Sleep Hours	7–8 hours/day
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Health Profile

Purpose: Understand medical background.

Medical Conditions

- Diabetes
- Hypertension
- Thyroid Disorders
- PCOS

- Kidney Disease
- Liver Disease
- Heart Disease
- Anemia
- Other (Custom)

Allergies

- Milk
- Eggs
- Peanuts
- Seafood
- Gluten
- Soy
- None
- Other

Medications (Optional)

- Current medications
- Vitamin supplements

Dietary Preferences

Purpose: Generate personalized meal plans.

- Vegetarian
- Vegan
- Eggetarian
- Non-Vegetarian
- Jain Diet
- Keto

- Low Carb
- High Protein

Food Preferences

- Favorite foods
- Foods to avoid
- Religious restrictions

Health Goals

Purpose: Personalize recommendations.

Options:

- Weight Loss
- Weight Gain
- Muscle Gain
- Improve Immunity
- Improve Energy
- Improve Bone Health
- Healthy Lifestyle
- Maintain Current Weight

Daily Food Diary

This is one of the most important sections.

Breakfast

- Food Name
- Quantity
- Time

Lunch

- Food Name
- Quantity
- Time

Dinner

- Food Name
- Quantity
- Time

Snacks

- Food Name
- Quantity
- Time

Drinks

- Water Intake
- Tea/Coffee
- Juice
- Soft Drinks

Food Search Source

- USDA FoodData Central
- OpenFoodFacts
- IFCT (Indian Food Database)

Symptom Assessment

Instead of asking users to type symptoms, use a checklist with severity ratings.

Symptom

Severity (0–5)

Fatigue



Symptom	Severity (0–5)
---------	----------------

Hair Loss	★ ★
Dry Skin	★ ★ ★
Muscle Weakness	★ ★ ★ ★
Bone Pain	★ ★
Mood Changes	★ ★ ★
Difficulty Concentrating	★ ★
Pale Skin	★ ★ ★
Frequent Illness	★ ★
Slow Wound Healing	★ ★ ★
Poor Night Vision	★
Tingling in Hands/Feet	★ ★
Loss of Appetite	★
Brittle Nails	★ ★

Lifestyle Information

Purpose: Improve prediction accuracy.

- Smoking (Yes/No)
- Alcohol Consumption
- Exercise Frequency
- Daily Water Intake
- Daily Screen Time
- Stress Level
- Sleep Quality

Blood Test Reports (Optional)

Users can either upload a report or manually enter values.

Blood Parameters

Test	Unit
Hemoglobin	g/dL
Iron	µg/dL
Ferritin	ng/mL
Vitamin D	ng/mL
Vitamin B12	pg/mL
Calcium	mg/dL
Magnesium	mg/dL
Zinc	µg/dL
Blood Sugar	mg/dL
Cholesterol	mg/dL

Family History (Optional)

- Diabetes
- Heart Disease
- Obesity
- Thyroid Disorders
- Osteoporosis
- Cancer

Progress Tracking

Collected over time:

- Weekly Weight
- BMI Changes
- Daily Calories
- Water Intake
- Exercise Logs
- Deficiency Risk Score
- Meal Adherence
- Goal Progress

AI Model Input Features

These are the features your ML model will use:

- Age
- Gender
- Height
- Weight
- BMI
- Activity Level
- Medical Conditions
- Dietary Preference
- Food Diary Nutrients
- Symptom Severity
- Blood Test Values (if available)
- Lifestyle Factors
- Health Goals

AI Model Output

The model predicts the risk (e.g., Low, Medium, High or a probability score) for deficiencies such as:

- Iron
- Vitamin D
- Vitamin B12
- Calcium
- Protein
- Vitamin C
- Folate
- Magnesium
- Zinc
- Potassium

It can also provide:

- Confidence Score
- Contributing Factors (via SHAP)
- Recommended Foods
- 7-Day Meal Plan
- Daily Nutrient Targets

Recommended User Flow

Register/Login



Complete Health Profile

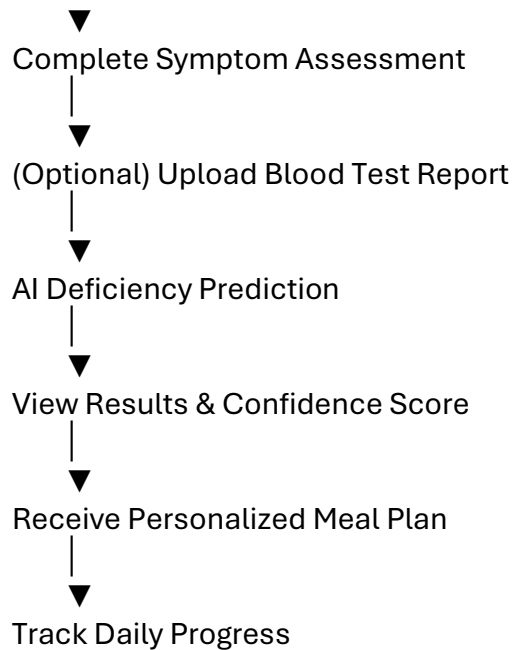


Set Health Goals



Record Food Diary





16. Detailed GitHub Repository Structure

AI-Based-Nutritional-Assessment-System/

```
|
|
|— README.md          # Project overview and setup instructions
|— LICENSE            # Open-source license
|— .gitignore         # Ignore unnecessary files
|— .env.example        # Sample environment variables
|— docker-compose.yml  # Docker configuration (optional)
|— requirements.txt    # Python dependencies
|
|— docs/              # Project Documentation
|   |— Project_Proposal.pdf
|   |— SRS.pdf
```

```
| |─ Software_Design_Document.pdf
| |─ API_Documentation.md
| |─ Database_Design.md
| |─ Architecture.md
| |─ Deployment_Guide.md
| |─ User_Manual.md
| |─ Testing_Report.md
| |─ Sprint_Plan.md
| |─ Presentation/
| |   |─ Final_PPT.pptx
| |   |─ Demo_Script.md
|
|─ frontend/          # React Application
| |
| |─ public/
| |   |─ favicon.ico
| |   |─ logo.png
| |   |─ images/
| |
| |─ src/
| |
| |   |─ assets/
| |       |─ icons/
| |       |─ images/
| |       |─ fonts/
| |       |
```



```
| | └─ components/
| | |
| | | └─ common/
| | | | └─ Button.jsx
| | | | └─ Input.jsx
| | | | └─ Card.jsx
| | | | └─ Loader.jsx
| | | | └─ Modal.jsx
| | | | └─ Alert.jsx
| | | |
| | | └─ layout/
| | | | └─ Navbar.jsx
| | | | └─ Sidebar.jsx
| | | | └─ Footer.jsx
| | | | └─ DashboardLayout.jsx
| | | |
| | | └─ forms/
| | | | └─ RegisterForm.jsx
| | | | └─ LoginForm.jsx
| | | | └─ HealthProfileForm.jsx
| | | | └─ FoodDiaryForm.jsx
| | | | └─ SymptomsForm.jsx
| | | | └─ BloodReportForm.jsx
| | | |
| | | └─ charts/
| | | | └─ NutritionChart.jsx
```

| | | └─ ProgressChart.jsx

| | | └─ CaloriesChart.jsx

| | | └─ RiskChart.jsx

| | |

| | └─ pages/

| | | └─ Home/

| | | └─ Login/

| | | └─ Register/

| | | └─ Dashboard/

| | | └─ HealthProfile/

| | | └─ FoodDiary/

| | | └─ Symptoms/

| | | └─ BloodReport/

| | | └─ Prediction/

| | | └─ MealPlanner/

| | | └─ Reports/

| | | └─ Profile/

| | | └─ Settings/

| | | └─ Admin/

| | |

| | └─ services/

| | | └─ authService.js

| | | └─ userService.js

| | | └─ foodService.js

| | | └─ predictionService.js

```
| | | └─ mealService.js
| | | └─ dashboardService.js
| | |
| | └─ hooks/
| | └─ context/
| | └─ routes/
| | └─ utils/
| | └─ constants/
| | └─ App.jsx
| | └─ main.jsx
| |
| └─ package.json
| └─ vite.config.js
|
└─ backend/
| |
| └─ app/
| |
| | └─ api/
| | | └─ auth.py
| | | └─ users.py
| | | └─ profile.py
| | | └─ food_diary.py
| | | └─ symptoms.py
| | | └─ blood_reports.py
| | | └─ prediction.py
```

```
| | | ├── recommendations.py
| | | ├── dashboard.py
| | | └── admin.py
| | |
| | | ├── auth/
| | | │   ├── jwt_handler.py
| | | │   ├── hashing.py
| | | │   └── permissions.py
| | |
| | | ├── config/
| | |
| | | ├── database/
| | | │   ├── connection.py
| | | │   ├── session.py
| | | │   └── migrations/
| | |
| | | ├── models/
| | | │   ├── user.py
| | | │   ├── profile.py
| | | │   ├── food_diary.py
| | | │   ├── symptom.py
| | | │   ├── blood_report.py
| | | │   ├── prediction.py
| | | │   ├── meal_plan.py
| | | │   └── progress.py
| | |
```

```
| | ├── schemas/
| | ├── services/
| | ├── middleware/
| | ├── utils/
| | |
| | ├── ml/
| | |   ├── predictor.py
| | |   ├── preprocessing.py
| | |   ├── recommendation_engine.py
| | |   ├── model_loader.py
| | |   └── explainability.py
| | |
| | └── main.py
| |
| └── requirements.txt
|
└── ml/
    |
    | ├── datasets/
    | |   ├── raw/
    | |   ├── processed/
    | |   └── external/
    | |
    | |
    | ├── notebooks/
    | |
    | └── preprocessing/
```

```
| | ├── clean_data.py
| | ├── feature_engineering.py
| | └── split_dataset.py
| |
| ├── training/
| | ├── random_forest.py
| | ├── train_model.py
| | └── evaluate_model.py
| |
| ├── saved_models/
| | ├── nutrition_model.pkl
| | ├── label_encoder.pkl
| | └── scaler.pkl
| |
| └── reports/
|     ├── model_accuracy.pdf
|     ├── confusion_matrix.png
|     └── feature_importance.png
|
| ├── database/
| | ├── schema.sql
| | ├── seed.sql
| | ├── migrations/
| | └── ER_Diagram.png
|
| ├── tests/
```

- | |└─ backend/
- | |└─ frontend/
- | |└─ api/
- | |└─ integration/
- | |└─ performance/
- |
- |└─ postman/
- | |└─ Nutrition_API.postman_collection.json
- | |└─ Environment.json
- |
- |└─ deployment/
- | |└─ Dockerfile.backend
- | |└─ Dockerfile.frontend
- | |└─ nginx.conf
- | |└─ render.yaml
- |
- |└─ scripts/
- | |└─ setup.sh
- | |└─ seed_database.py
- | |└─ create_admin.py
- | |└─ deploy.sh
- |
- |└─ .github/
- | |└─ workflows/
- | | |└─ backend-ci.yml
- | | |└─ frontend-ci.yml

```
| | └─ tests.yml
| |
| | └─ ISSUE_TEMPLATE/
| |   └─ bug_report.md
| |   └─ feature_request.md
| |   └─ task.md
| |
| | └─ PULL_REQUEST_TEMPLATE.md
| └─ CODEOWNERS
|
└─ .vscode/
    └─ settings.json
    └─ extensions.json
    └─ launch.json10
```